



CUBAN NATIONAL CHEMISTRY CONTEST

"the future of our country must necessarily be a future of men of sciences, must be a future of men of thinking, because that is precisely what we are creating..."

Fidel Castro

Name(s): _____ Last name(s): _____

School: _____

Municipality: _____ Province: _____

CODE: _____ Test: **DAY 2. TWELFTH GRADE EXAM**

Instructions

- Write clearly all your data in the appropriate place (names, last names, school, municipality, province). Do not write anything in the field CODE.
- Verify that your questions booklet is complete. This is the Day 2 problem set, with an individual exam for 12th grade. The exam has 2 problems (problem 10, 8 items; problem 11, 12 items) and a total of 5 pages.
- All data, constants, conversions and basic equations needed are included in the exam. You are allowed to use a scientific calculator.
- At the beginning of each problem its scoring (from a total of 100 points), each item scoring (in marks) is indicated.

Success!



Ministerio de Educación
República de Cuba



CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's Constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gasses' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	Thermodynamics First Principle	$Q = \Delta E + W$
Cero in the Celsius scale	273,15 K	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{C_{\text{red}}}{C_{\text{ox}}}$
Water ionic product at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$
Standard conditions	$p^\circ = 100 \text{ kPa}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$	$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$		$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$	

PERIODIC TABLE OF THE ELEMENTS

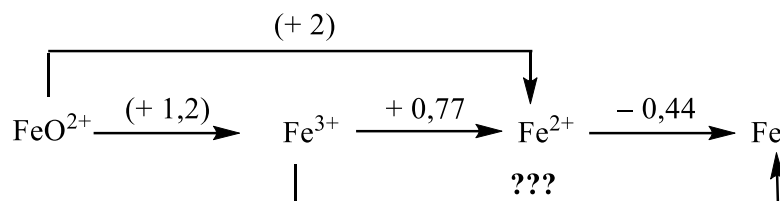
PERIODIC TABLE OF THE ELEMENTS																		
1 1A												13 3A		14 4A	15 5A	16 6A	17 7A	18 8A
1 H 1.008	2 2A											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18	
3 Li 6.941	4 Be 9.012											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95	
11 Na 22.99	12 Mg 24.31	3 3B	4 4B	5 5B	6 6B	7 7B	8 8B	9 8B	10 8B	11 1B	12 2B	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80	
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3	
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)	
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6		114 Uuq (277)		116 Uuh (277)		118 Uuo (277)	
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (262)	108 Hs (265)	109 Mt (266)	110 Ds (269)	111 Rg (272)	112 Uub (277)							
		58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0			
		90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)			

Problem 10. Electrochemistry (20 points)

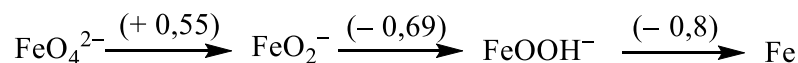
10.1	10.2	10.3	10.4	10.5	10.6	10.7	10.8	Total	Total
4	4	4	4	6	6	6	6	40	20

The following Latimer diagram corresponds to iron species in acid and basic media.

Acid media



Basic media



10.1. In which media is Fe(III) more oxidant? In which media metallic Fe is more reductant?

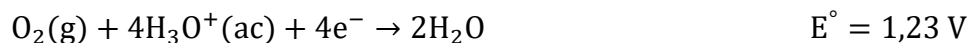
10.2. Does any of the iron species disproportionate? Justify.

10.3. May species Fe(III) and Fe coexist simultaneously in an acid solution? Justify numerically.

10.4. Calculate the standard potential $E^\circ(\text{Fe}^{3+}/\text{Fe})$.

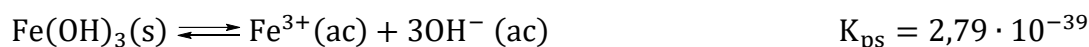
It is possible to understand the redox processes occurring in natural waters with the aid of electrochemical data. For example, in a lake, cold in the depth and warmer on the surface, the temperature gradients avoid vertical mixing. On the surface water is completely oxygenated and iron is present in the form of insoluble particles of $\text{Fe}(\text{OH})_3$. These particles tend to sink. More profound, the content of O_2 is low. If the organic content and other sources of reducing agent are high enough, the hydroxide will be reduced, and the iron will dissolve in the form of Fe^{2+} . Then, the formed ions will diffuse to the surface, where they will be in contact with O_2 and oxidize again to $\text{Fe}(\text{OH})_3$.

10.5. Estimate the approximate potential on the surface of an ordinary lake. Consider that the temperature is 25 °C, the pressure is 1,01 bar and the pH is 7,0. Air composition is N_2 78%, O_2 21% and Ar 1% (volume percent).

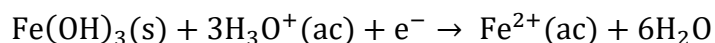


In fact, natural water conditions are slightly different from those calculated by you due to the presence of several species in the solution.

10.6. Calculate Fe^{3+} and Fe^{2+} concentrations on the surface of the lake. Consider that the temperature is 25 °C, the pH is 6,5 and the potential is 0,65 V.



10.7. Demonstrate that the standard potential corresponding to the following solution process that occurs close to the bottom of the lake is $E^\circ = 0,97 \text{ V}$.



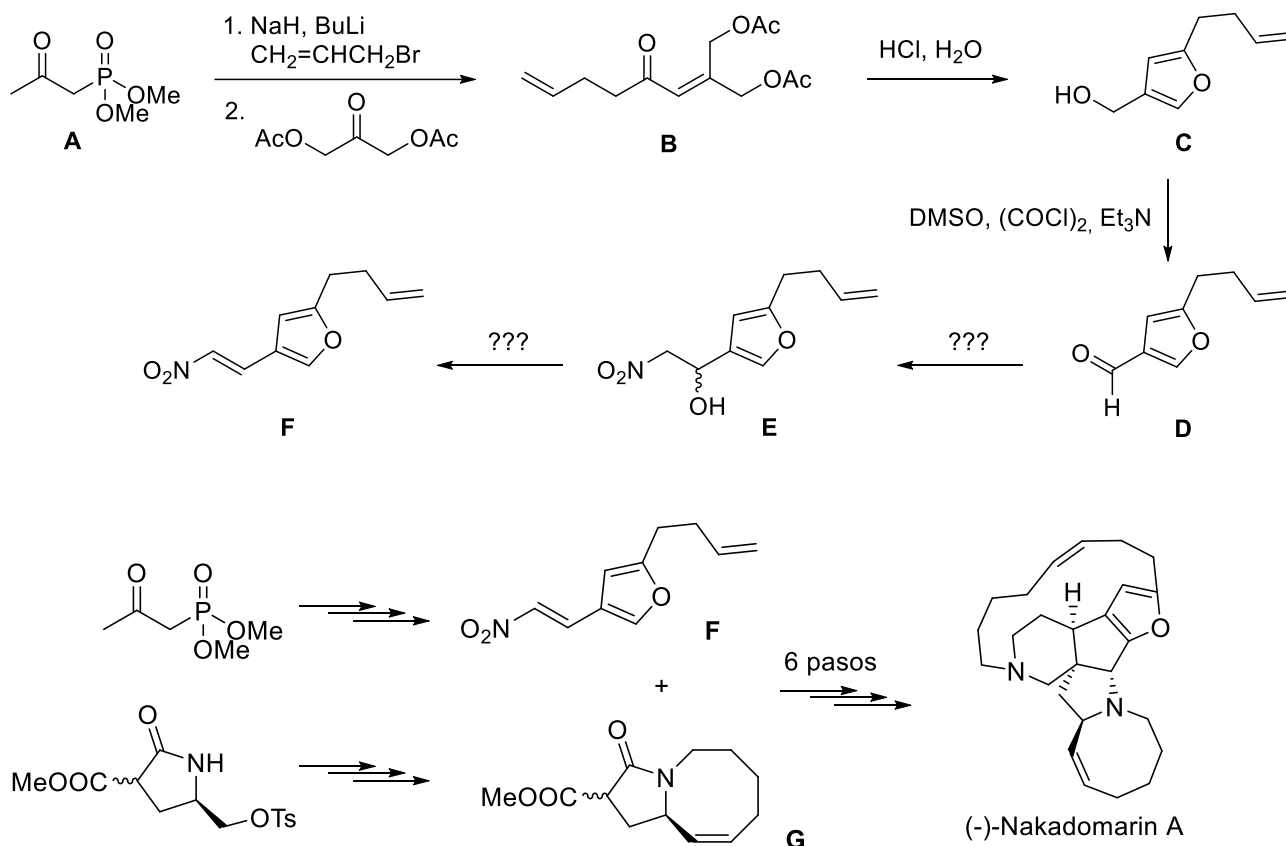
10.8. Calculate Fe^{3+} and Fe^{2+} concentrations at the bottom of the lake. Consider that the temperature is 15 °C, the pH is 6,0 and the potential is 0,15 V. Overlook the effects of the temperature on the equilibrium constant and the standard potential.

Problem 11. Organic Chemistry (30 points)

10.1	10.2	10.3	10.4	10.5	10.6	10.7	10.8	10.9	10.10	10.11	10.12	Total	Total
6	8	4	3	4	4	3	3	11	7	4	3	60	30

The carbonyl group in aldehydes and ketones is one of the most versatile groups from a synthetic point of view. Therefore, numerous compounds used as intermediaries in complex synthetic routes have this chemical function.

(-)-Nakadomarin A is a member of manzamines, alkaloids isolated from marine sponges, which possess a unique hexacyclic structure. The following scheme shows a fragment of the total synthesis of this alkaloid that was carried out by Darren J. Dixon and coworkers (Jakubec, P.; Cockfield, D. M.; Dixon, D. J., J. Am. Chem. Soc. 2009, 131, 16632–16633).



11.1. Propose a mechanism for the preparation of **B** from **A**.

11.2. Propose a mechanism for the preparation of **C** from **B**.

11.3. The obtaining of **C** is favored for forming an aromatic furan ring. Mention three characteristics a system must have to be considered aromatic.

11.4. What is the function of the mixture DMSO, (COCl)₂, Et₃N: oxidizing agent, reducing agent, solvent, dehydrating or deprotection mixture? DMSO = dimethylsulfoxide

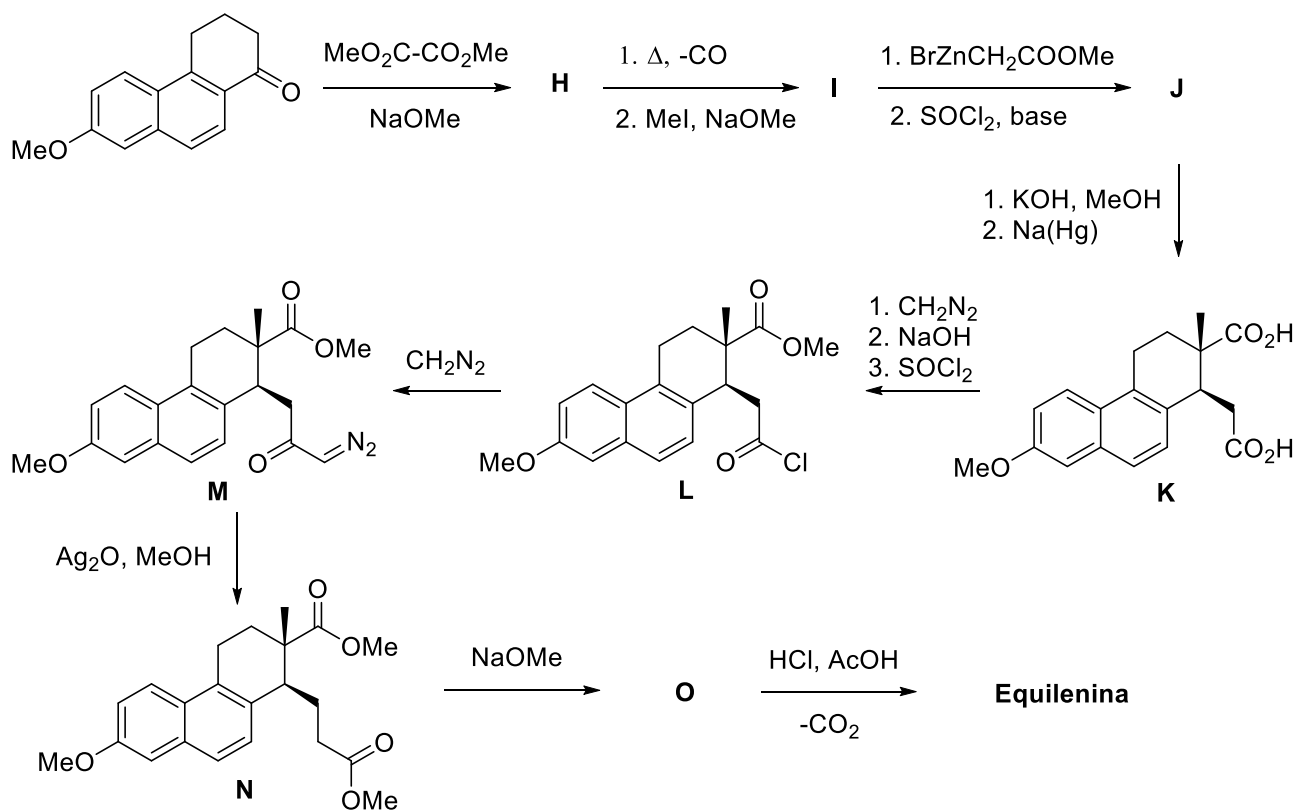
11.5. Suggest reagents for the conversion of **D** to **E**.

11.6. Suggest reagents that allow the conversion of **E** to **F**.

11.7. What is the name of the chemical reaction happening when intermediates **F** and **G** are mixed in the presence of a base?

11.8. Which is the maximum number of stereoisomers that (-)-*Nakadomarin A* can have?

Equilenine is a steroidal sexual hormone from the family of estrogens. It was the first natural product of a complex structure that was completely synthesized. The scheme corresponding to the total synthesis of equilenine (Bachmann, W. E.; Cole, W.; Wilds, A. L., J. Am. Chem. Soc. 1940, 62 (4), 824–839) is given below.



11.9. Represent the structures of compounds **H**, **I** and **J**.

11.10. Represent the structures of compounds **O** and *Equilenine*.

11.11. During the transformation of α -diazoketone **M** catalyzed by Ag(I), a very reactive intermediate **X** is formed, which quickly adds methanol giving **N**; **X** also reacts quickly with water, alcohols and amines. Represent the structure of **X**.

11.12. Diazomethane (CH₂N₂) is a reagent widely employed in organic synthesis. **Suggest** what is the objective of the previously described synthetic step.